Yijun YUAN (袁 轶君)

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Research Interests

- **p-adic Galois representations**
- **p-adic transcendental number theory**
- **Formalization of arithmetic geometry**

Employment

2025–2027 **Postdoctoral Fellow, Institute for Theoretical Sciences, Westlake University**
Advisor: Huayi Chen

Education

2022–2025 **Ph.D., Yau Mathematical Sciences Center, Tsinghua University**
Pure Mathematics
Thesis title: *The Field of Norms and $(\varphi, \Gamma_{\mathfrak{F},K})$-Modules for p-Adic False Tate Curve Extensions.*
Advisor: Lei Fu / Haoran Wang

2018 – 2021 **M.Sc., Shanghai Center for Mathematical Sciences, Fudan University**
Pure Mathematics
Thesis title: *Uniformizer of the False Tate Curve Extension of $\mathbb{Q}_p$.*
Advisor: Shanwen Wang / Quanshui Wu

2014 – 2018 **B.S., School of Mathematical Sciences, East China Normal University**
Mathematics and Applied Mathematics

Research Publications

Journal Articles

1. S. Wang and **Y. Yuan**, "Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$ (II)," *International Journal of Number Theory*, vol. 20, no. 01, pp. 1–18, 2024. DOI: 10.1142/S1793042124500015
2. S. Wang and **Y. Yuan**, "Uniformizer of the false Tate curve extension of $\mathbb{Q}_p$," *The Ramanujan Journal*, vol. 58, no. 2, pp. 549–595, 2022. DOI: 10.1007/s11139-021-00473-0

Preprints (all submitted)

1. **Y. Yuan**, *Descent of (φ, τ)-modules in characteristic p*, 2025. arXiv: 2511.17029 [math.NT].
2. S. Wang and **Y. Yuan**, *On the p-adic transcendence of $\sum_{k=1}^{\infty} p^{-1/p^k}$*, 2025. arXiv: 2509.24609 [math.NT].
3. **Y. Yuan**, *Formalization of Harder-Narasimhan theory*, 2025. arXiv: 2509.19632 [math.AG].

4. **Y. Yuan**, *Multivariable period rings of p-adic false tate curve extension*, 2025. arXiv: 2505.24064 [math.NT].
5. S. Wang and **Y. Yuan**, *Hyper-algebraic invariants of p-adic algebraic numbers*, 2024. arXiv: 2402.15947 [math.NT].
6. S. Wang and **Y. Yuan**, *Truncated expansion of ζ_{p^n} in the p-adic mal'cev-neumann field*, 2023. arXiv: 2111.07127 [math.NT].

Talks

2025, Nov. 19th — **Formalization of Harder-Narasimhan Theorey**
Shanghai Institute for Mathematics and Interdisciplinary Sciences

2025, Jun. 13th — **Arithmetic of spherically complete valued fields in mixed characteristic**
Symposium on Number Theory and Algebraic Geometry of Spring 2025
Capital Normal University

2024, Nov. 1st — **Hyper-algebraic invariants of p-adic algebraic numbers**
Second Doctoral Conference on Frontiers of Mathematics
School of Mathematical Sciences, Peking University

2024, Oct. 16th — **Arithmetic of p-adic Mal'cev-Neumann field**
Number Theory Launch Seminar
Beijing Institute of Mathematical Sciences and Applications (BIMSA)

2023, Sep. 27th — **(φ, Γ)-modules for p-adic false Tate curve extension**
0-1 Seminar
Department of Mathematical Sciences, Tsinghua University

Awards

2024 — Outstanding Teaching Assistant in Undergraduate Fundamental Courses, Tsinghua University (Second Class)

2023 — Scholarship of Yau Mathematical Sciences Center (Second Class)

Tsinghua University Comprehensive Excellence Scholarship (Second Class)

Peer Review

2024 — Journal of Number Theory

Teaching Experience

2025, Autumn — TA for **Foundation of Algebra and Analysis** at Westlake University.

2024, Autumn — TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

2023, Autumn — TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

2022, Autumn — TA for an undergraduate course on **Linear Algebra** at Tsinghua University.

Skills

Coding — Python (SageMath), LaTeX, Lean4, Mathematica, …